

Manav Vora

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EDUCATION

University of Illinois Urbana Champaign (UIUC)

Aug 2022 – Present

Ph.D. candidate in Aerospace Engineering (Robotics and AI Focus)

GPA: 3.9

Advisor: Prof. Melkior Ornik

Outstanding Academic & Research, Department Fellowship

Indian Institute of Technology Bombay (IITB)

Aug 2018 – Aug 2022

Bachelors of Technology in Aerospace Engineering

CGPA: 4.0 (Minors GPA: 4.0)

Minors in Systems and Controls Engineering

Honors Degree (Department rank: 3)

SUMMARY

Research in scalable, resource-aware multi-agent decision-making and robotics, integrating meta-learning, deep reinforcement learning, and generative modeling for intelligent, interpretable autonomy.

TECHNICAL SKILLS

Programming Languages: Python; C++; Julia; MATLAB; Bash

ML Frameworks: PyTorch; TensorFlow; AIMET; MLFlow; StableBaselines3; Gym

Robotics/Sim: Robot Operating System (ROS/ROS2); OpenCV; Gazebo; SUMO; Simulink

LLM Systems: LangGraph; LangChain; ChromaDB

RESEARCH EXPERIENCE

LEADCAT (LEarning, Decision, Control, AuTonomy) Lab, UIUC

Aug 2022 – Present

Research Assistant

- Scalable multi-agent RL under partial observability for resource- and capacity-constrained systems. (*pub:* [1], [7], [8])
- Learning-based planning and control under uncertainty, including meta-trained RL+MPC and belief-aware diffusion for real-time trajectory planning. (*pub:* [2], [4])
- Advanced assured autonomy: reachability-guided planning for unknown nonlinear dynamics and verified collision-avoidance wrappers for learned controllers. (*pub:* [3], [10])
- Scalable coordination and routing methods for modular agents on graphs using force-based interaction models and optimization. (*pub:* [6])
- LLM safety and alignment via continual superalignment framing under evolving human norms. (*pub:* [9])
- Scalable communication policies for cooperative multi-agent RL through latent grouping. (*pub:* [5])
- Multi-modal multi-robot search-and-rescue via diffusion world models and RL. (*ongoing*)
- Long-tail joint trajectory forecasting under severe occlusions via structured priors + tail-aware residual learning for rare behaviors. (*ongoing*)

Corteva

Aug 2025 – Dec 2025

Industry-Sponsored Research

MAESTRO: Multi-Agent Expert System for Transparent Response Orchestration in Supply Chain Problems

- Built a multi-agent chatbot that delivers transparent, source-grounded explanations of supply-chain optimization outputs for non-technical stakeholders.

- Implemented an agentic RAG workflow in LangGraph with iterative retrieve–tool-use–validate loops; routed queries to role-based expert agents and synthesized answers via an aggregator.
- Added a Pandas/Excel analysis node to compute and inject data facts (KPIs, throughput summaries) from solver spreadsheets for quantitative questions.
- Open-sourced the system and delivered a stakeholder-facing report and evaluation artifacts. Code available at <https://github.com/Manavvora/MAESTRO>

Rivian

May 2025 – Aug 2025

Machine Learning Intern

Quantization-Aware Optimization of Multi-Modal Motion Prediction Models

- Built a scalable, multi-modal multi-agent motion prediction model extending HPTR and Wayformer, trained and benchmarked on the Argoverse 2 Motion Forecasting dataset.
- Designed an INT16 post-training quantization pipeline achieving $<0.01\%$ mAP degradation and $<0.1\%$ minADE increase relative to FP32 with $2\times$ reductions in memory and compute cost.
- Proposed and implemented a mixed-precision INT8+INT16 quantization scheme that reduced computation by $\approx 12\%$ while limiting accuracy loss to $\approx 0.02\%$, providing an optimal trade-off for edge deployment.
- Developed layer-wise calibration tools to automatically identify precision-sensitive modules and enable targeted recovery during quantization.

GRASP Lab, UPenn

Apr 2021 – Feb 2022

Research Intern

Neural Network-Based ODEs and Lie Algebra for Robotic Control Systems

- Achieved 95% of the performance of RRT-star on actual dynamics by leveraging RRT-star with Lie Algebra Neural-ODE generated dynamics.
- Investigated Hamiltonian-based Neural ODEs for pendulum and quadrotor systems, leveraging underlying system structure.

HONORS AND AWARDS

Cozad New Venture Challenge (2026): Third place; secured **\$125,000** total in venture support, including follow-on investments from Multimodal Ventures and Pear VC.

CoRL 2025: Selected among the top 5% of accepted papers for oral presentation.

IROS 2024: Selected among the top 10% of accepted papers for oral presentation.

IEEE CDC Student Travel Award

UIUC Aerospace Engineering Department Fellowship (2023) for research & academic achievements

Chanakya Research Fellowship (2021) for autopilot design for delivery drones.

PUBLICATIONS AND PREPRINTS

1. **Vora, M.**, P. Thangeda, M. N. Grussing, and M. Ornik. 2023. Welfare Maximization Algorithm for Solving Budget-Constrained Multi-Component POMDPs. *IEEE Control Systems Letters* 7: 17361741.
2. Puthumanaim, G. *, **M. Vora** *, and M. Ornik. 2024. ComTraQ-MPC: Meta-Trained DQN-MPC Integration for Trajectory Tracking with Limited Active Localization Updates. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. **Oral presentation (Top 5%)**
3. Z. Zhang, G. Puthumanaim, **M. Vora**, and M. Ornik. 2025. Motion Planning and Control with Unknown Nonlinear Dynamics through Predicted Reachability. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. **Oral presentation (Top 5%)**
4. G. Puthumanaim et. al. 2025. “Belief-Conditioned One-Step Diffusion: Real-Time Trajectory Planning with Just-Enough Sensing.” *Conference on Robot Learning (CoRL)*. **Oral presentation (Top 5%)**

5. **Vora, M.**, G. Puthumanaim, H. Tsukamoto, M. Ornik. 2026. “SCoUT: Scalable Communication via Utility-Guided Temporal Grouping in Multi-Agent Reinforcement Learning.” *arXiv preprint* arXiv:2603.04833. (Under Review at Reinforcement Learning Conference (RLC))
6. A. Casselman, **M. Vora**, and M. Ornik. 2025. “Virtual Force-Based Routing of Modular Agents on a Graph.” *arXiv preprint* arXiv:2505.00928. (Under Review)
7. **Vora, M.**, J. Liang, M. N. Grussing, and M. Ornik. 2025. Solving Truly Massive Budgeted Monotonic POMDPs with Oracle-Guided Meta-Reinforcement Learning. *arXiv preprint* arXiv:2408.07192. (Under review at Transactions on Machine Learning Research (TMLR))
8. **Vora, M.**, I. Shomorony, and M. Ornik. 2025. Capacity-Aware Planning and Scheduling in Budget-Constrained Multi-Agent MDPs: A Meta-RL Approach. *IEEE Robotics and Automation Letters (RAL)*
9. Puthumanaim, G. *, **M. Vora** *, P. Thangeda, and M. Ornik. 2024. A Moral Imperative: The Need for Continual Superalignment of Large Language Models. *arXiv preprint* arXiv:2403.14683.
10. Puthumanaim, G. *, **M. Vora** *, T. Shafa, Y. Li, M. Ornik, and S. Mitra. 2024. Assured Collision Avoidance for Learned Controllers: A Case Study of ACAS Xu. In *AIAA SCITECH 2024 Forum*, 1168.

*Equal contribution.

RESEARCH PROJECTS

Pixels → Prompts → Pose: Building (and Breaking) a VLA+RL Policy

- Integrated TinyVLA and DrQ-v2 Soft ActorCritic to fine-tune vision-language-action (VLA) policies for 6-DoF robotic manipulation under natural-language prompts.
- Introduced a three-stage pipeline combining visionlanguage embedding alignment, LoRA-based low-rank RL fine-tuning, and latent probing for interpretability.
- Deployed on a real SO-100 manipulator to diagnose reward-hacking and specification-gaming behaviors, improving robustness through concept-probe diagnostics.

Manifold-Aware Latent Diffusion Models

- Proposed a geometrically consistent latent diffusion framework that learns a symmetric positive-definite Riemannian metric to respect data manifold curvature.
- Developed a three-stage training schemeVAE pretraining, metric-network learning, and manifold-aware diffusionto achieve locally isometric latent representations.
- Demonstrated improved geodesic interpolation and sample quality on Swiss-Roll and MNIST datasets, achieving smoother transitions and geometry-preserving generations.

Vision-based Spacecraft Trajectory Estimation

- Developed a methodology for spacecraft trajectory estimation using 2D images captured by onboard cameras to reconstruct 3D spatial paths, focusing on the OSIRIS-REx mission’s Departure Flyby phase.
- Implemented Structure from Motion (SfM) techniques, integrating SIFT for feature detection and RANSAC for outlier mitigation, and evaluated trajectory estimation using global and incremental SfM via OpenMVG, achieving high precision compared to actual mission trajectory data.

TEACHING, MENTORSHIP AND SERVICE

- Teaching Assistant for AE 352: Dynamical Systems with 60+ students. **Jan-May 2024, 2025**
- Mentoring 4 undergraduate students conducting research in multi-agent systems.
- Reviewer: **L4DC** (2026), **ICLR** (2025), **IEEE RAL** (2025), **ICRA** (2025, 2024), **IROS** (2025, 2024), **ITSC** (2025), **CDC** (2024), **ACC** (2024), **IEEE LCSS** (2024).